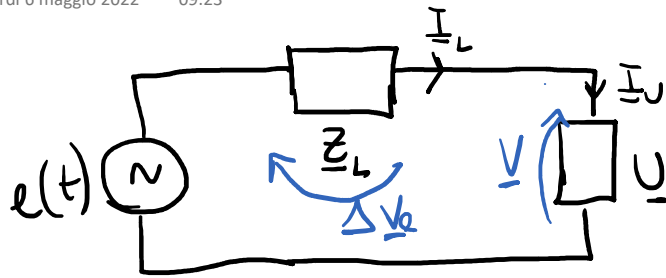


RIFASAMENTO IN MONOFASE

venerdì 6 maggio 2022 09:23



$\underline{Z}_L$: impedenza di linea

$$= R_L + j\omega L$$

$\underline{U}$: utilizzatore

$$(\underline{R}_L - L)$$

$$\underline{I}_L = \underline{I}_U$$

- P_{de} : perdite per effetto Joule lungo la linea

$$P_{de} = R_L \cdot I_e^2 \propto I_e^2$$

$$- \underline{V} = \underline{E} - \underline{Z}_L \underline{I}_e \propto I_e$$

Pot. ass. dal carico : $P = V I_U \cos \varphi = V I_e \cos \varphi$

$$\Rightarrow I_e = \frac{P}{V \cos \varphi}$$

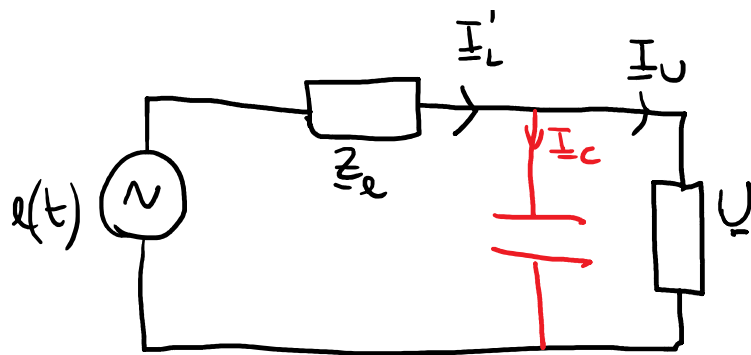
Ridurre I_e :

- $P \downarrow$ NO

- $V \uparrow$ PROBLEMI DI ISOLAMENTO NO

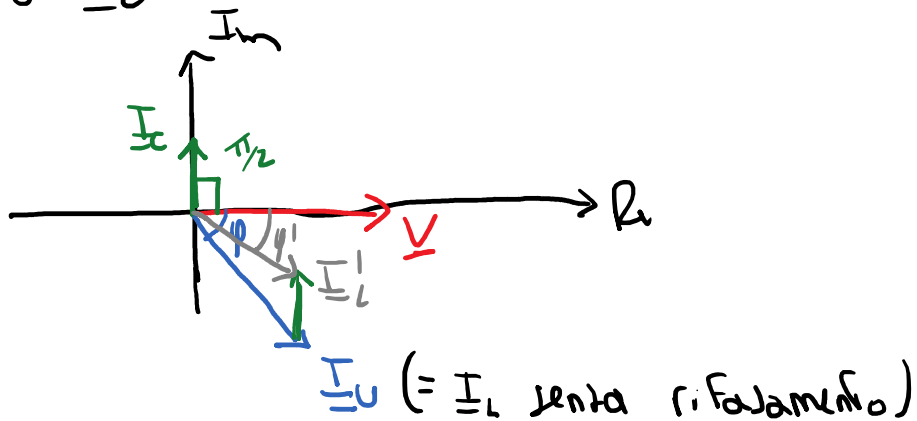
$\cos \varphi \uparrow \rightarrow \varphi \downarrow \Rightarrow$ meno più in fase $\underline{V}$ e $\underline{I}$

$\Downarrow$
RIFASAMENTO

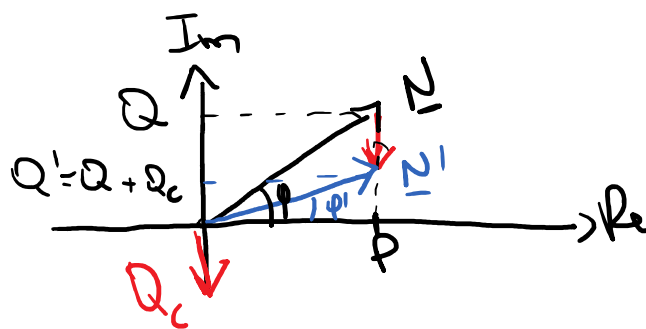


U è $\omega L - L$
 INSERIAMO IN
 PARALLELO AL CARICO
 UN CONDENSATORE

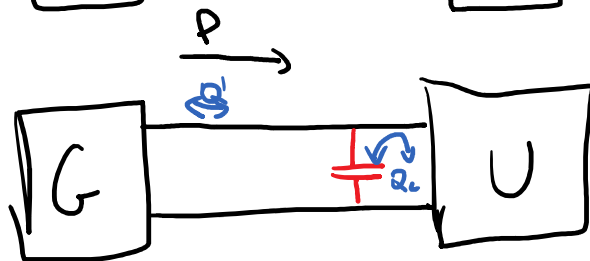
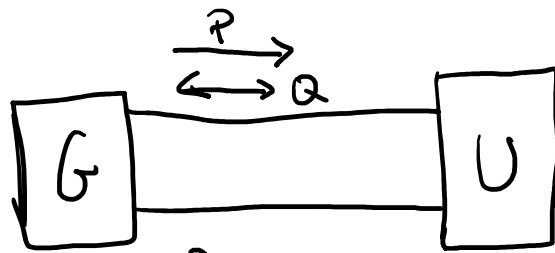
$$\underline{I}_L' = \underline{I}_U + \underline{I}_C$$



$$\varphi' < \varphi \Rightarrow \cos \varphi' > \cos \varphi \Rightarrow |\underline{I}_L'| < |\underline{I}_L| \Rightarrow P_{de} \downarrow \Delta V_L \downarrow$$



$$\frac{P}{\dots}$$



$$Q' = Q + Q_c$$

$$\cos \varphi = 0,95$$

Quanto deve valere C?

$$\begin{cases} Q = P \tan \varphi \\ Q' = Q + Q_c = P \tan \varphi' \end{cases}$$

$$Q' - Q = Q + Q_c - Q = Q_c = P(\tan \varphi' - \tan \varphi)$$

$$Q_c = X_c \cdot I^2 = -\frac{1}{\omega C} \cdot I^2 = -\frac{1}{\omega C} (\omega^2 C^2 V^2) \quad \begin{matrix} \nearrow I = \omega C V \end{matrix}$$

$$Q_c = -\omega C V^2 = P(\tan \varphi' - \tan \varphi)$$

$$\Rightarrow C = \frac{P(\tan \varphi - \tan \varphi')}{\omega V^2}$$

$$\omega_p \geq 0,95 \rightarrow \text{OK}$$

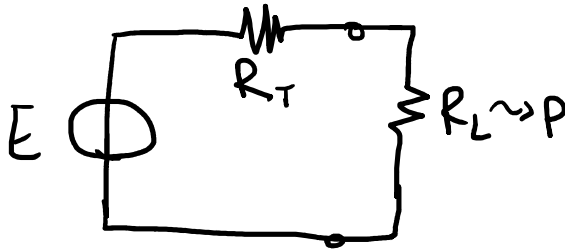
$$0,1 \leq \omega_p < 0,95$$

$$\omega_p < 0,1 \rightarrow \text{obblig. di R.F.}$$

MASS. TRASF. DI POTENZA

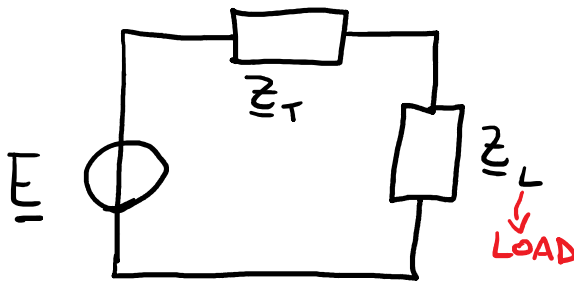
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IN DC:



$$P = P_{max} \quad R_L = R_T$$

IN AC:



$$Z_L = ? \text{ affinché}$$

$$P = P_{max}$$

$$Z_T = R_T + j X_T$$

$$Z_L = R_L + j X_L$$

$$\underline{I} = \frac{\underline{E}}{\underline{Z}_T + \underline{Z}_L} = \frac{\underline{E}}{(R_T + R_L) + j(X_T + X_L)}$$

$$P = R_L I^2$$

$$|\underline{I}| = \frac{|\underline{E}|}{|\underline{Z}_T|} = \frac{E}{\sqrt{(R_T + R_L)^2 + (X_T + X_L)^2}}$$

$$P = R_L \cdot \frac{E^2}{(R_T + R_L)^2 + (X_T + X_L)^2}$$

$$\frac{\partial P}{\partial X_L} = 0 \Rightarrow \frac{\partial P}{\partial X_L} = -R_L \cdot E^2 \frac{2(X_T + X_L)}{((\dots)^2 + (\dots)^2)} = 0$$

$$\Rightarrow \boxed{X_L = -X_T}$$

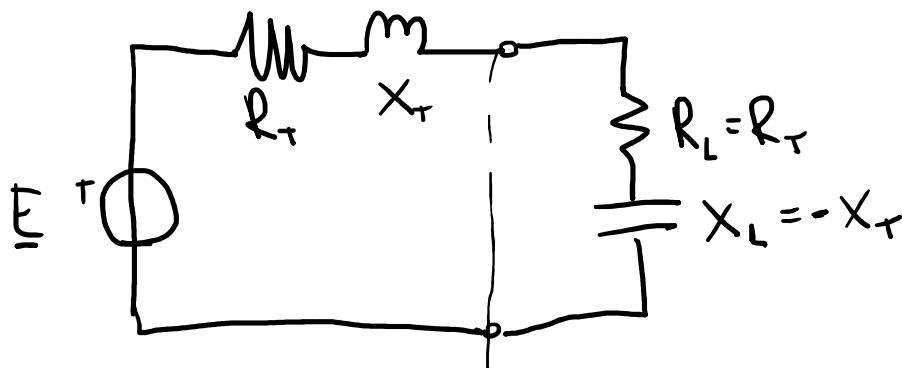
$$\frac{\partial P}{\partial R_L} = 0 \Rightarrow \frac{\partial P}{\partial R_L} = \frac{E^2 [(R_T + R_L)^2 + \cancel{(X_T + X_L)^2}] - R_L E^2 2(R_T + R_L)}{(\dots)^4} = 0$$

$$E^2 (R_T + R_L)^2 - R_L E^2 2(R_T + R_L) = 0$$

$$(R_T + R_L) [(R_T + R_L) - 2R_L] = 0$$

$$\hookrightarrow \boxed{R_L = R_T}$$

$$P = P_{max} \quad \text{se} \quad \underline{Z}_L = \underline{Z}_T^* = R_T - jX_T$$



TEOR. DI BOUCHEROT

venerdì 6 maggio 2022 10:39

Teor. di Tellegen:

$$\sum_{m=1}^{N_g} P_m(t) = \sum_{k=1}^{N_{s.i.}} P_k(t)$$

generatori: utilizzatori:

$$\sum_{m=1}^{N_g} P_m = \sum_{k=1}^{N_{s.i.}} P_k$$

$$\sum_{m=1}^{N_g} (P_m + jQ_m) = \sum_{k=1}^{N_{s.i.}} (P_k + jQ_k)$$

$$\sum_{m=1}^{N_g} P_m = \sum_{k=1}^{N_{s.i.}} P_k$$

$$\sum_{m=1}^{N_g} Q_m = \sum_{k=1}^{N_{s.i.}} Q_k$$